

“What is a Living Thing”

Reference Definitions and Formal Conditions accompanying the paper at <https://projectvitalis.substack.com/publish/post/185074781>

Dr. Sean O’Grady

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1. Base Definitions

A thing is a bounded composition of matter in space that is distinct from its surroundings.

$A(x)$ = a thing that is alive

$B(x)$ = boundary of a thing

t = time

2. Boundary Definition

The boundary of a thing may strengthen, remain stable, or degrade over time. This behavior is captured by the time derivative of the boundary function.

$d \frac{B(x)}{d} t \geq 0$, where

$d \frac{B(x)}{d} t = 0$, boundary stable

$d \frac{B(x)}{d} t > 0$, boundary becoming more defined

$d \frac{B(x)}{d} t < 0$, boundary becoming less defined

3. Entropy Definitions

Entropy within and across a boundary can be decomposed into internal accumulation, external export, and total internal and external components.

S_{acc} = entropy accumulating inside the boundary

S_{exp} = entropy exported from inside to outside the boundary

S_i = total entropy inside the boundary

S_o = entropy outside the boundary

4. Entropy Gradient Statements

The internal entropy gradient of a thing is determined by the balance between entropy accumulation and entropy export across its boundary.

$$d \frac{S_i}{d} t = \frac{d}{d} t, (S_{\text{acc}} - S_{\text{exp}})$$

$d\frac{S_i}{dt} > 0$, entropy increases internally, reducing the gradient“

$d\frac{S_i}{dt} < 0$, entropy decreases internally, maintaining the gradient“

5. Boundary Maintenance Definitions

Boundary maintenance describes how usable energy is converted into restored order to preserve the integrity of a thing over time.

O_B = order within the boundary

E_B = usable free energy allocated to boundary maintenance, expressed as a portion of G_{use}

C_B = conversion efficiency of usable energy into restored order

f_{B_M} = function that maintains the boundary, in units of order per unit time

D_B = natural decay rate, in units of order per unit time

6. Boundary Maintenance Function

The boundary maintenance function expresses the rate at which usable energy is converted into restored boundary order.

$f_{B_M} = C_B, E_B$ (boundary order per unit time)

7. Boundary Order Dynamics

The boundary order changes over time as maintained order competes with natural decay.

$d\frac{O_B}{dt} = f_{B_M} - D_B$

8. Energy and Instructions Definitions

These quantities extend the entropy framework to include usable energy and internal instruction sets that support boundary maintenance and repair.

G_{use} = usable free energy that can be captured for self-directed work

I_{int} = internal set of instructions for boundary repair

9. Conditions for Living Things

A(x) is true if and only if all of the following conditions hold.

$\frac{d}{dt}(S_{exp} - S_{acc}) > 0$, entropy export exceeds entropy accumulation

$f_{B_M} > D_B$, boundary maintenance rate exceeds boundary decay rate

$G_{use} > 0$, the entity captures and uses free energy for self-directed work

$I_{int} = \text{true}$, internal instruction set for boundary repair is present

10. Case Study – Boulder

Applying the conditions for living things to the boulder.

$\frac{d}{dt}(S_{exp} - S_{acc}) < 0$, entropy accumulation exceeds entropy export

$G_{\text{use}} = 0$, the system captures energy in the form of heat but cannot use it to do work“
 $f_{B_M} < D_B$, the boundary is not maintained by the system faster than it decays
 $I_{\text{int}} = \text{false}$, no internal instruction set for directing boundary repair is present

The boulder is not alive.

11. Case Study – Barred Owl

Applying the conditions for living things to the barred owl.

$\frac{d}{dt}(S_{\text{exp}} - S_{\text{acc}}) > 0$, entropy export exceeds entropy accumulation
 $f_{B_M} > D_B$, boundary maintenance rate exceeds boundary decay rate
 $G_{\text{use}} > 0$, the owl captures and uses free energy for self-directed work
 $I_{\text{int}} = \text{true}$, internal instruction set for boundary repair is present

The barred owl is alive, but not in the same way as the owl.

12. Case Study – Mouse

Applying the conditions for living things to the mouse.

$\frac{d}{dt}(S_{\text{exp}} - S_{\text{acc}}) > 0$, entropy export exceeds entropy accumulation
 $f_{B_M} > D_B$, boundary maintenance rate exceeds boundary decay rate
 $G_{\text{use}} > 0$, the mouse captures and uses free energy for self-directed work
 $I_{\text{int}} = \text{true}$, internal instruction set for boundary repair is present

The mouse is alive.

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With this, I also invite you to attempt other use cases to identify other cases that break the conditions for life that are proposed above. Interesting examples may include hurricanes, appliances, artificial intelligence, entire ecosystems, financial systems, and more. Please send all correspondence to projectvitalis@proton.me.